

Review Report

—by Koushik Das

Comparative Review: Seismic Hazard Assessment Frameworks and National Earthquake Monitoring Infrastructure in India

A comparative review of two papers (P1, P2) on India's seismic risk framework — one addressing hazard assessment methodology, the other addressing real-time earthquake monitoring infrastructure

1. Papers Reviewed

ID	Title & Authors	Venue / Year	Focus
P1	Seismic Hazard Assessment Frameworks in High-Risk Indian Regions: A Methodological Review and Gap Analysis with Reference to Itanagar, India — <i>Anshu et al.</i>	Natural Hazards Rev. (ASCE), 2026	State-of-the-art review of PSHA/DSHA methodology, site characterization, and ground response analysis, with Itanagar as a gap-analysis case study
P2	National Seismological Network in India for Real-Time Earthquake Monitoring — <i>Bansal et al.</i>	Seismol. Res. Lett. (SSA), 2021	Description of India's 115-station real-time monitoring network: instrumentation, data flow, catalog completeness, and dissemination

2. Overview

The two papers sit on opposite ends of the same problem: how India understands and responds to earthquake risk. **P1** is a methodological review of seismic *hazard assessment* — the frameworks (PSHA and DSHA), site characterization techniques (MASW, SPT, Vs30), and ground response models used to estimate how strongly a given location will shake in a future earthquake. It uses Itanagar as a case study to show where this analytical pipeline breaks down for under-studied cities. **P2**, on the other hand, describes the physical *monitoring infrastructure* — the 115-station National Seismological Network (NSN) that actually detects, locates, and reports earthquakes as they happen. Where P1 is concerned with predicting hazard before an earthquake, P2 is concerned with recording and disseminating information during and after one. Read together, they outline the full risk-management pipeline: P2 supplies the raw instrumental data (earthquake catalogues, magnitudes, locations) that P1's hazard assessment methods depend on — and P1 explicitly identifies incomplete instrumental catalogues as one of the central gaps limiting hazard studies in cities like Itanagar.

3. Comparison Matrix

Criteria	P1 — Hazard Assessment Review	P2 — Monitoring Network
Core subject	Methodology for estimating future seismic hazard (PSHA, DSHA, site response)	Infrastructure for recording earthquakes in real time as they occur
Type of paper	State-of-the-art / methodological review with regional gap analysis	Descriptive infrastructure and performance report
Geographic scope	India-wide review, focused case study on	All of India and adjoining regions (lat. 5–

Criteria	P1 — Hazard Assessment Review	P2 — Monitoring Network
	Itanagar, Arunachal Pradesh	40°N, long. 60–100°E)
Key data / metric	Vs30, PGA, G/Gmax and damping ratio curves, hazard curves	115 seismic stations; catalog of 34,000+ events since 1505; ~8 min average event-reporting time
Central finding	Itanagar (Zone V) lacks region-specific GMPEs, a complete catalogue, and nonlinear ground response modeling	The network has matured from 6 stations (1940) to 115 (2021), cutting event-location time from over an hour to ~8 minutes
Main limitation noted	Existing Itanagar hazard analysis borrows attenuation models from other regions due to data scarcity	Detection threshold still varies regionally (M 2.5 in Delhi vs. M 4.0 in border areas); network expansion still ongoing

4. Paper-wise Summary

P1 — Seismic Hazard Assessment Frameworks (Anshu et al., 2026)

This is a review, not a new applied study. It surveys the two dominant hazard assessment approaches — deterministic (DSHA), which evaluates a single worst-case earthquake scenario, and probabilistic (PSHA), which estimates the likelihood of exceeding various shaking levels over time. It walks through the full PSHA workflow: building and homogenizing an earthquake catalogue, declustering dependent events, characterizing seismic sources, applying ground-motion prediction equations (GMPEs), and generating hazard curves through a logic-tree treatment of uncertainty. It then covers site characterization (shear-wave velocity via MASW and SPT, Vs30-based NEHRP site classification) and ground response analysis, contrasting simpler equivalent-linear methods against more accurate but harder-to-implement nonlinear methods. Using Itanagar as its illustrative case, the paper argues that while cities like Delhi, Guwahati, and Chennai have undergone multiple rounds of hazard study, Itanagar — despite sitting in the highest-risk seismic zone (Zone V) — still lacks region-specific GMPEs, a complete earthquake catalogue, nonlinear ground response modeling, and dense geotechnical site investigation.

P2 — National Seismological Network in India (Bansal et al., 2021)

This paper documents the physical and operational side of Indian seismology: the National Seismological Network (NSN), maintained by the National Centre for Seismology (NCS). It traces the network's growth from a single observatory at Alipore in 1898 to 115 digital stations today, each equipped with a broadband seismometer and a strong-motion accelerograph, transmitting data in real time via VSAT to a central receiving station in New Delhi. The paper details the full data pipeline — from field acquisition, through SeisComp3 autolocation software, to manual review and public dissemination via SMS, app, and social media — and reports that the network now estimates earthquake source parameters within an average of 8 minutes, comparable to global networks such as USGS. It also presents a spatial analysis of the *magnitude of completeness* (the smallest earthquake magnitude the network reliably detects in a given area) across four time windows from 1898 to 2020, showing steady improvement nationwide, though detection thresholds still vary by region — as fine as M 2.5 in Delhi, but only M 4.0 in some border areas. The paper positions continued network expansion (150 stations planned, with 300 more envisioned) as key to further improving detection and response time.

5. Key Takeaways

- The two papers describe complementary halves of the same system: **P2's monitoring network is the data source that P1's hazard assessment methods rely on.** A better earthquake catalogue from the NSN directly addresses one of the central data gaps P1 identifies for cities like Itanagar.
- Both papers converge on the same weak point from different directions: **data completeness in India's northeast and border regions.** P2 shows detection thresholds are still coarser there (M 3.0–4.0) than in Delhi (M 2.5); P1 shows this directly limits how reliable hazard assessment can be for a Zone V city like Itanagar.
- P1 is forward-looking and analytical (what *could* happen, and how strongly), while P2 is operational and infrastructural (what *did* happen, and how fast we knew about it). Neither is complete without the other for effective earthquake risk management.
- Both papers use the same underlying catalogue-processing concepts — magnitude homogenization (converting Mb, Ms, ML to a common Mw scale) and the **magnitude of completeness (Mc)** — and even reference the same software tool, ZMAP, for computing it, underscoring how tightly the two research areas are methodologically linked.
- Neither paper alone gives a full picture of seismic risk in a specific city: P1 has the hazard methodology but flags Itanagar's data as incomplete, while P2 has the national data infrastructure but does not translate raw detections into localized hazard or design guidance.

6. Conclusion & Future Directions

Together, these two papers sketch the full arc of seismic risk management in India — from detecting an earthquake as it happens (P2) to assessing, ahead of time, how badly a given location will be shaken by one (P1). P2 demonstrates that India's instrumental capacity has improved dramatically, with reporting times now competitive with global networks. P1 demonstrates that this improved capacity has not yet been fully translated into localized hazard products for many high-risk, understudied cities such as Itanagar. Future work would benefit from more directly linking the two: using the NSN's expanding, region-specific earthquake catalogues to build region-specific GMPEs and complete the seismicity parameters that PSHA studies for cities like Itanagar currently have to borrow from elsewhere. As both papers separately suggest, continued network densification (P2) and continued methodological refinement, including nonlinear ground response modeling and Bayesian PSHA (P1), point toward the same conclusion: seismic risk management in India is an ongoing, data-hungry process, not a one-time deliverable.